

4.3 Common Properties of the Determinant

S

Core skills you'll master in this section:

- ▶ When a square matrix's column vectors are collinear or coplanar, the area or volume is 0, meaning the square matrix is not invertible.
- ▶ The determinant of a product of square matrices equals the product of their individual determinants.
- ▶ Swapping any two rows or two columns flips the sign of the determinant.
- ▶ If every row and column of a square matrix is simultaneously multiplied by a scalar k , the determinant scales by a factor of k^n .
- ▶ A permutation matrix is used to swap rows or columns; it is an orthogonal matrix, and its determinant is ± 1 .
- ▶ Transposing a square matrix does not change its determinant.

With the geometric intuition for determinants built up in the previous two sections, understanding the common properties of determinants introduced in this section becomes intuitive and natural. Next, let's use specific examples to explain these properties.

A Determinant of 0

There are several situations in which a square matrix's determinant is 0; let's go over them below.

The following 2×2 matrix's column vectors are collinear, so its determinant is 0

$$\det \begin{pmatrix} 1 & 3 \\ 1 & 3 \end{pmatrix} = 0$$

As shown in Figure 1 (a), $\mathbf{a}_1$ and $\mathbf{a}_2$ lie on the same line, so the parallelogram spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$ has area 0.

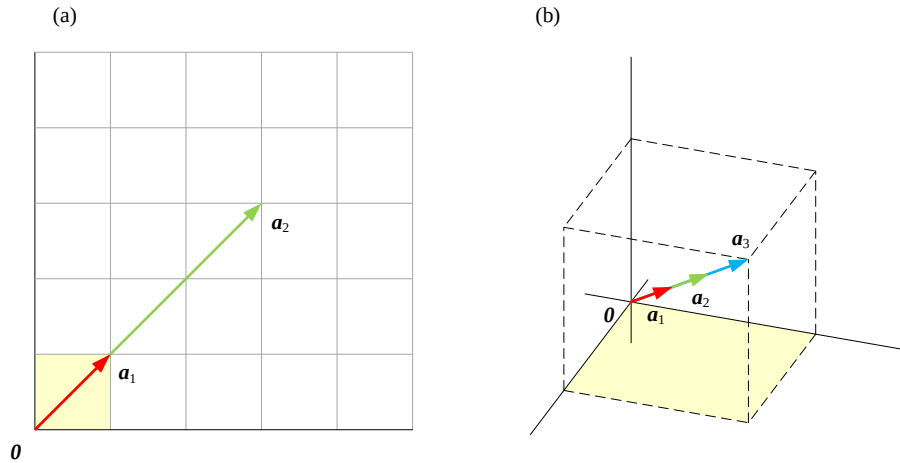


Figure 1. A Matrix's Column Vectors Are Collinear

The following 3×3 matrix's column vectors are collinear, so its determinant is 0

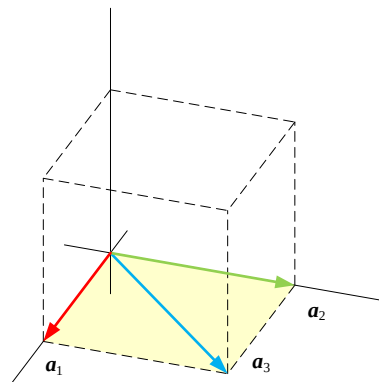
$$\det \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = 0$$

As shown in Figure 1 (b), in three-dimensional space, the three column vectors of the matrix above are collinear, so they cannot span a parallelepiped, and its volume is 0.

The determinant of the following 3×3 matrix is also 0

$$\det \begin{pmatrix} 3 & 0 & 3 \\ 0 & 3 & 3 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

As shown in Figure 2, in three-dimensional space, the matrix's three column vectors in the equation above are coplanar, so they likewise cannot span a parallelepiped, and its volume is also 0.

Figure 2. A 3×3 Matrix's Column Vectors Are Coplanar

This also tells us that if a matrix has an entire row or column of 0s, the determinant is 0. For example

This PDF file is the author's draft, published to make it convenient for readers to study on mobile devices; the final content is subject to the printed edition published by Tsinghua University Press.

Copyright belongs to Tsinghua University Press. Please do not use it commercially, and please credit the source when quoting.

Code and PDF file downloads: <https://github.com/Visualize-ML>

The companion video lectures for this book are all published on Bilibili — 生姜 DrGinger: <https://space.bilibili.com/513194466>

Comments and corrections are welcome. The book's dedicated email address: jiang.visualize.ml@gmail.com

$$\det\begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix} = 0$$

Another example

$$\det\begin{pmatrix} 0 & 0 \\ 2 & 2 \end{pmatrix} = 0$$

To summarize, when square matrix $\mathbf{A}$'s determinant is 0, this means the square matrix's column vectors are linearly dependent.

The Determinant of a Product of Matrices

You're already well aware that the determinant can be viewed as the scaling factor for area (or volume).

If one square matrix $\mathbf{A}$ scales area (or volume) by a factor of $\det(\mathbf{A})$, and another square matrix $\mathbf{B}$ scales area (or volume) by a factor of $\det(\mathbf{B})$, then the combined scaling factor is the product of the two.

This means that the determinant of the matrix product $\mathbf{AB}$ equals the determinant of square matrix $\mathbf{A}$ times the determinant of square matrix $\mathbf{B}$, that is

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

The above equation also tells us that,

$$\det(\mathbf{BA}) = \det(\mathbf{B}) \cdot \det(\mathbf{A})$$

This tells us that order affects the result of the transformation ($\mathbf{AB} \neq \mathbf{BA}$), even though the area doesn't change.

For example, given the following 2×2 square matrices $\mathbf{A}$ and $\mathbf{B}$

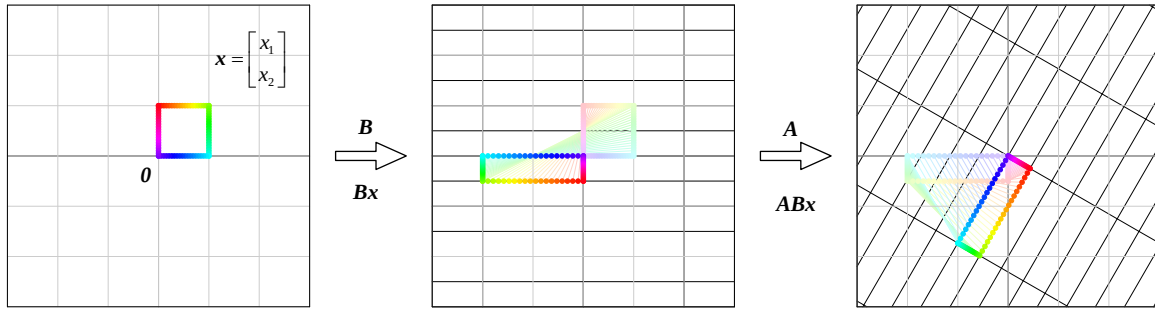
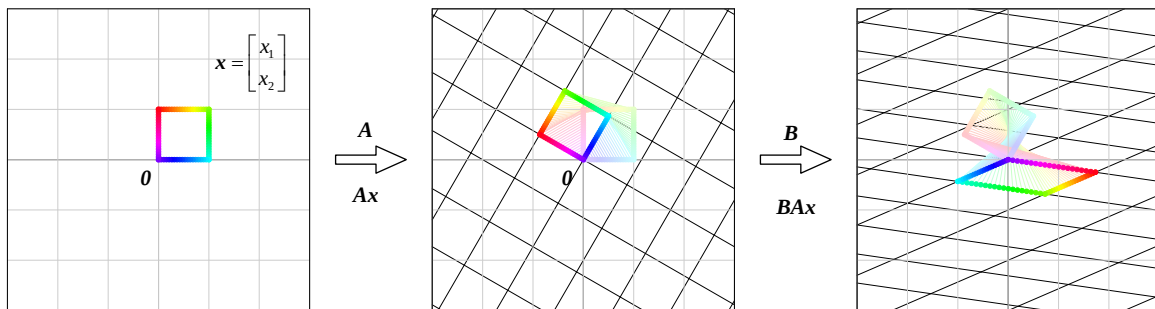
$$\mathbf{A} = \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -2 & 0 \\ 0 & -1/2 \end{bmatrix}$$

The effect of square matrix $\mathbf{A}$ is to rotate the plane around the origin; the effect of square matrix $\mathbf{B}$ is "scaling + mirroring."

It's easy to compute that square matrices $\mathbf{A}$ and $\mathbf{B}$'s determinants are both 1, that is, $\det(\mathbf{A}) = \det(\mathbf{B}) = 1$.

Figure 3 shows the geometric transformation corresponding to the matrix multiplication $\mathbf{ABx}$. The column vectors $\mathbf{x}$ correspond to the unit square, with area 1; after the $\mathbf{AB}$ transformation, the area is still 1.

Figure 4 corresponds to $\mathbf{BAx}$'s geometric transformation; clearly, Figure 3 is different from Figure 4. But Figure 4's final geometric shape still has area 1.

Figure 3. $ABx = y$'s Corresponding Geometric TransformationFigure 4. $BAx = y$'s Corresponding Geometric Transformation

Swapping Two Columns or Rows Flips the Sign of the Determinant

If a square matrix's determinant is nonzero, swapping any two of its columns flips the sign of the determinant. Swapping any two of its rows also flips the sign of the determinant.

This is easiest to understand using the permutation matrix. A permutation matrix is a special kind of square matrix obtained by permuting the rows or columns of the identity matrix. A permutation matrix is an orthogonal matrix.

A permutation matrix's entries consist only of 0s and 1s; each row and each column of the matrix has exactly one 1, and all other entries are 0.

For example, the 2×2 permutation matrix P

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

For matrix A , AP 's effect is to swap A 's columns 1 and 2

$$AP = \begin{bmatrix} a & b \\ c & d \end{bmatrix} @ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} @ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix}$$

PA 's effect is to swap A 's rows 1 and 2

$$PA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} @ \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} c & d \\ a & b \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

Compute square matrix P 's determinant, and you'll find $\det(P) = -1$.

Let's look at two slightly more complex examples. Matrix A times the permutation matrix changes the order of A 's column vectors.

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} = \begin{bmatrix} a_{1,3} & a_{1,1} & a_{1,4} & a_{1,2} \\ a_{2,3} & a_{2,1} & a_{2,4} & a_{2,2} \\ a_{3,3} & a_{3,1} & a_{3,4} & a_{3,2} \\ a_{4,3} & a_{4,1} & a_{4,4} & a_{4,2} \end{bmatrix}$$

Similarly, left-multiplying matrix $\mathbf{A}$ by this permutation matrix changes $\mathbf{A}$'s row-vector ordering

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \\ \mathbf{a}^{(3)} \\ \mathbf{a}^{(4)} \end{bmatrix} = \begin{bmatrix} \mathbf{a}^{(2)} \\ \mathbf{a}^{(1)} \\ \mathbf{a}^{(4)} \\ \mathbf{a}^{(3)} \end{bmatrix}$$

Multiplying a Matrix by a Scalar

For an $n \times n$ square matrix $\mathbf{A}$, the scalar multiplication $k\mathbf{A}$'s determinant is

$$\det(k\mathbf{A}) = k^n \det(\mathbf{A})$$

From a geometric perspective, the above equation is easy to understand.

For example, given the 2×2 matrix $\mathbf{A}$, $k\mathbf{A}$ can be written as

$$k\mathbf{A}_{2 \times 2} = \begin{bmatrix} k & \\ & k \end{bmatrix} \mathbf{A}_{2 \times 2}$$

Geometrically speaking, the above equation tells us that k scales the geometric shape's area by a factor of k^2 .

For example, in Figure 5, scalar 2 enlarges the area of the planar shape by a factor of 4. As shown in Figure 6, scalar 2 enlarges the volume of the three-dimensional shape by a factor of 8.

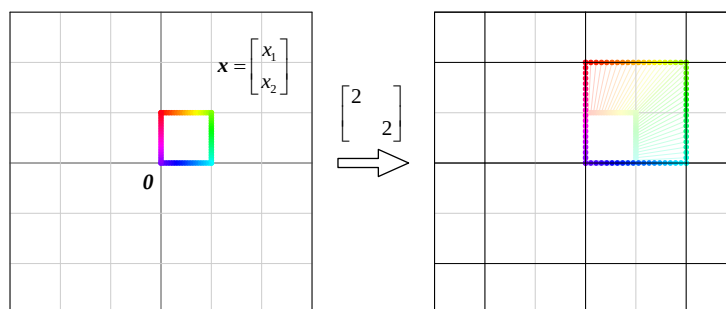
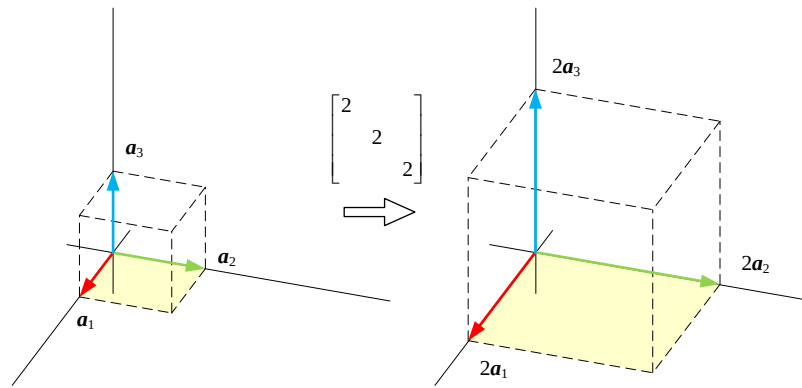


Figure 5. Given the 2×2 Matrix $\mathbf{A}$, $2\mathbf{A}$'s Geometric Transformation Corresponding to Scalar 2

Figure 6. Given the 3×3 Matrix A , $2A$'s Geometric Transformation Corresponding to Scalar 2

Transposing Doesn't Change the Determinant's Value

Square matrix A 's determinant equals its transpose A^T 's determinant

$$\det(A) = \det(A^T)$$

Explaining the above equation from a geometric perspective requires singular value decomposition (SVD). Simply put, SVD decomposes square matrix A into

$$A = USV^T$$

Here, U is an orthogonal matrix, representing a rotation. U 's determinant has absolute value 1.

S is a diagonal square matrix, representing a scaling; clearly, $S = S^T$.

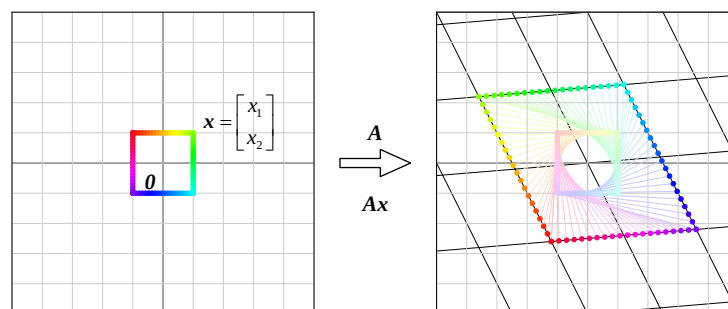
V is an orthogonal matrix, representing a rotation; V^T is also an orthogonal matrix, and also represents a rotation. V 's determinant has absolute value 1.

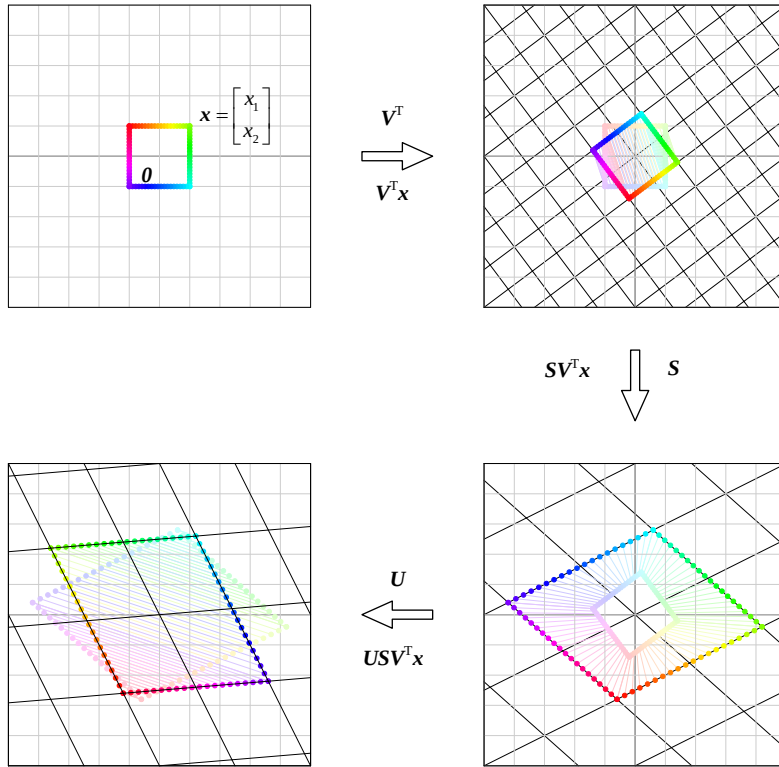
Let's look at an example below. Given the following matrix A

$$A = \begin{bmatrix} -1.2 & -2.4 \\ 2.4 & -0.2 \end{bmatrix}$$

Computing A 's determinant gives $\det(A) = 6$.

Figure 7 shows the "one-step" geometric transformation that matrix A produces on the column vectors x . If we replace matrix A with USV^T , we can obtain a step-by-step geometric transformation, specifically as shown in Figure 8.

Figure 7. $Ax = y$'s One-Step Geometric Transformation

Figure 8. $USV^T x = y$'s Step-by-Step Geometric Transformation

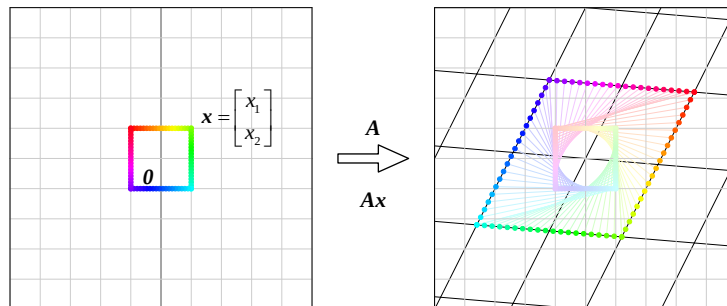
Square matrix A 's transpose can then be written as

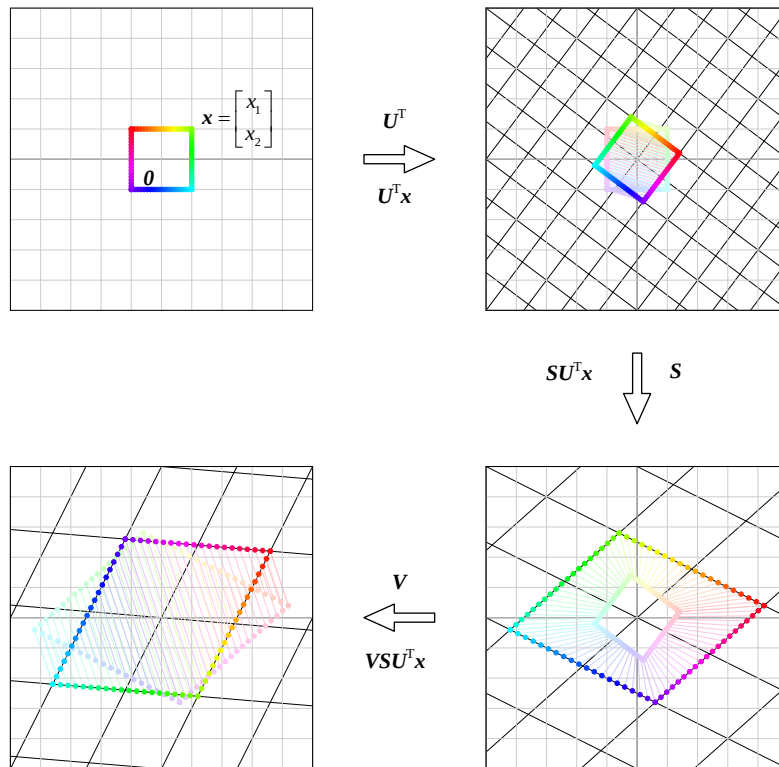
$$(A)^T = (USV^T)^T = VS^T U^T = VSU^T$$

From a geometric perspective, in matrix A , the factor that truly affects the shape's area or volume comes from the diagonal matrix S .

This tells us that U and V produce the same change in area or volume.

Figure 9 shows the “one-step” geometric transformation that matrix A^T produces on the column vectors x . If we replace matrix A^T with VSU^T , we can obtain a step-by-step geometric transformation, specifically as shown in Figure 10.

Figure 9. $A^T x = y$'s One-Step Geometric Transformation

Figure 10. $VSU^T x = y$'s Step-by-Step Geometric Transformation

For now, you only need a preliminary impression of SVD; SVD is a topic this book will cover in dedicated detail later on.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please determine whether the following determinants are 0.

► $\begin{bmatrix} 1 & -2 \\ 2 & -4 \end{bmatrix}$

► $\begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

► $\begin{bmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Q2. Given an arbitrary 2×2 matrix A , and the following diagonal matrix D , please explain what geometric transformations AD and DA undergo.

► $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

▶ $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$

Q3. Given an arbitrary 3×3 matrix $\mathbf{A}$, and the following permutation matrix $\mathbf{P}$, please explain what geometric transformations $\mathbf{AP}$ and $\mathbf{PA}$ undergo.

▶ $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

Q4. Please study singular value decomposition on your own.

Q5. Please learn to use the following NumPy function to perform an SVD decomposition of .

<https://numpy.org/doc/2.2/reference/generated/numpy.linalg.svd.html>